

I'm going to take the approach of building up the BLE devices from the ground up, so I'll need to start by creating characteristics, then services, then the BLE device for the ESP32.

```
pause = BLEBoolCharacteristic(uuid, BLERead | BLEWrite | BLENotify)
skipTrack = BLEBoolCharacteristic(uuid, BLERead | BLEWrite | BLENotify)
prevTrack = BLEBoolCharacteristic(uuid, BLERead | BLEWrite | BLENotify)
volUp = BLEBoolCharacteristic(uuid, BLERead | BLEWrite | BLENotify)
volDown = BLEBoolCharacteristic(uuid, BLERead | BLEWrite | BLENotify)
```

Media Controls

```
mediaControls = BLEService(uuid)
mediaControls.addCharacteristic(pause)
mediaControls.addCharacteristic(skipTrack)
mediaControls.addCharacteristic(prevTrack)
mediaControls.addCharacteristic(volUp)
mediaControls.addCharacteristic(volDown)
```

512 & 20 limit.

```
songName = BLECharacteristic(uuid, BLERead | BLEWrite, " ")
artistName = BLECharacteristic(uuid, BLERead | BLEWrite, " ")
songDuration = BLEUnsignedShortCharacteristic(uuid, BLERead | BLEWrite)
time = BLEUnsignedShortCharacteristic(uuid, BLERead | BLEWrite)
cover = BLECharacteristic(uuid, BLERead | BLEWrite, 512)
```

← Theoretically max size is 512 bytes, but maybe 20, in which case I'll need to use a buffer.

```
mediaMetadata = BLEService(uuid)
mediaMetadata.addCharacteristic(songName)
mediaMetadata.addCharacteristic(artistName)
mediaMetadata.addCharacteristic(songDuration)
mediaMetadata.addCharacteristic(time)
mediaMetadata.addCharacteristic(cover)
```

Now all of the necessary services & characteristics should exist, so I can start to set up the BLE device.

```
BLE.begin()
BLE.setLocalName("esp32 project")
BLE.setDeviceName("esp32 project")
BLE.addService(mediaControls)
BLE.addService(mediaMetadata)
while (BLE.connected()) {
    main
},
```